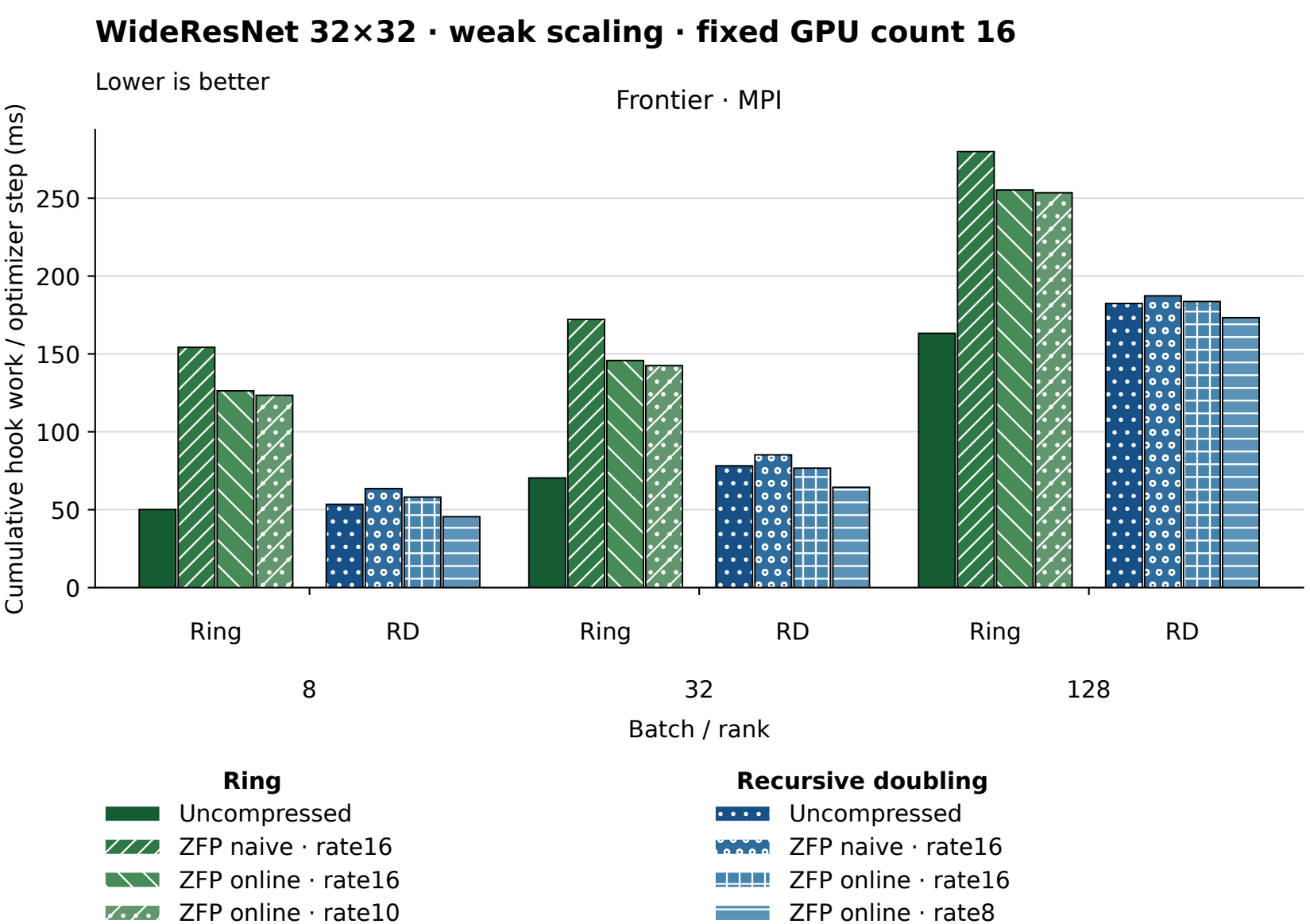

WideResNet 32×32 · weak scaling · fixed GPU count 16
Lower is better
Frontier · MPI
Cumulative hook work / optimizer step (ms)
0
50
100
150
200
250
Ring
RD
Ring
RD
Ring
RD
8
32
128
Batch / rank
Ring
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate10
Recursive doubling
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate8
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate